EXERCISE 4 - The Impact of Data Heterogeneity

1. What is Non-IID Data Distribution?

Non-IID (Non-Independent and Identically Distributed) data distribution in federated learning refers to a scenario where the data across different clients (participants) is not independently and identically distributed. This means that:

- **Data is not independent:** The data samples across clients are correlated or dependent on each other
- **Data is not identically distributed:** Different clients have different data distributions, patterns, or characteristics

Aspect	**IID Distribution**	**Non-IID Distribution**
Data Distribution	Same across all clients	Different across clients
Class Distribution	Balanced across clients	Skewed or missing classes
Convergence	Generally faster	Slower, may not converge
Real-world Applicability	Rare in practice	Common in real scenarios

Privacy	Lower privacy risk	Higher privacy risk

2. Pathological

A pathological split is a specific method for creating highly non-IID data distributions in federated learning. It's designed to simulate realistic scenarios where clients have access to only a limited subset of classes, creating extreme data heterogeneity.
Key Characteristics

- **Class Restriction**: Each client receives data from only a small number of classes (typically 1-2 classes)
- **Extreme Non-IID:** This creates the most challenging non-IID scenario possible
- **Realistic Simulation:** Mimics real-world situations where clients have specialized or limited data

Example with MNIST
For MNIST (10 classes) with 5 clients and 2 classes per client:
Total shards: 5 clients × 2 classes = 10 shards
Each shard: Contains samples from exactly one class

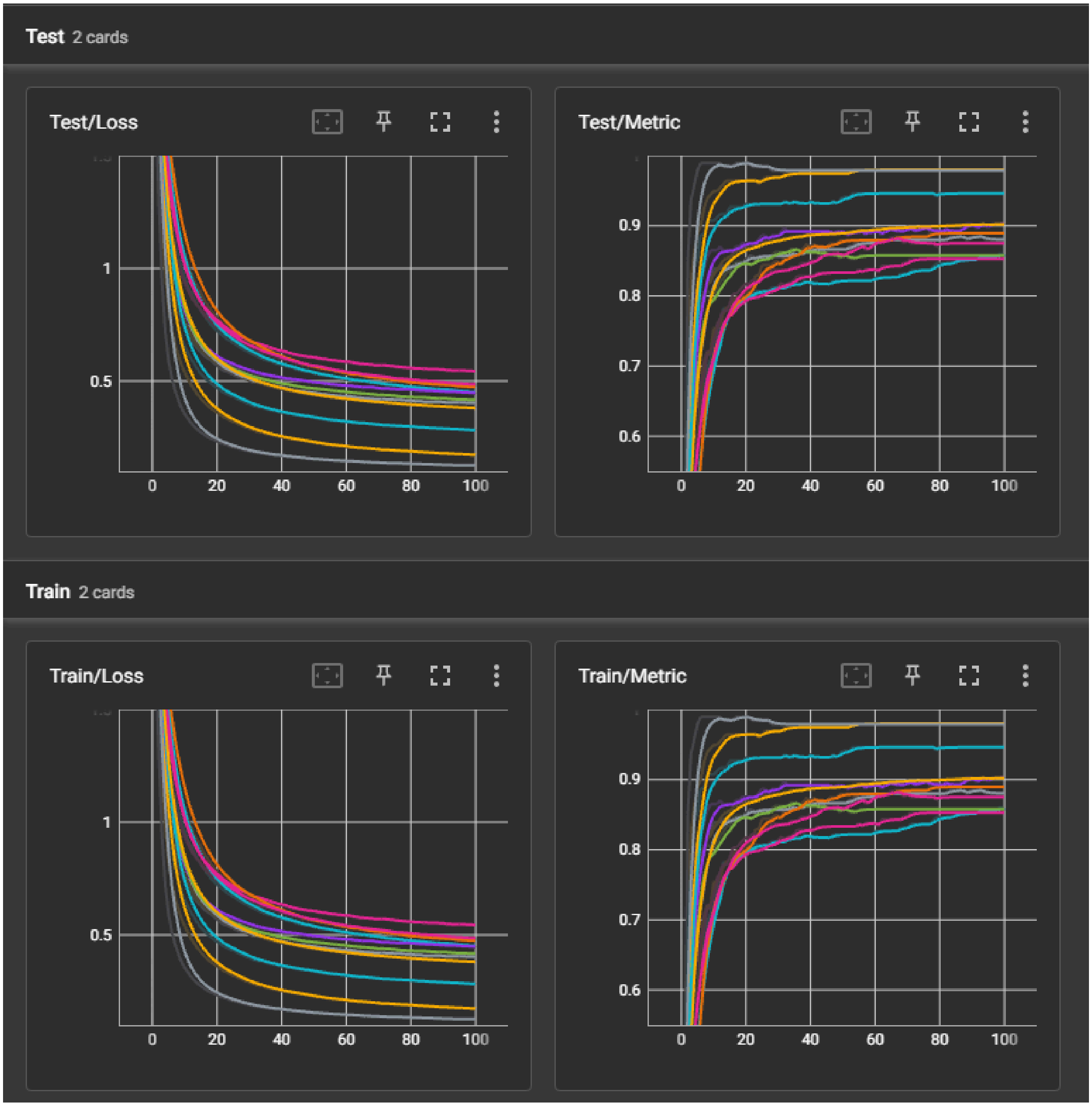
- Client 0: Gets 2 shards (e.g., classes 0 and 1)
- Client 1: Gets 2 shards (e.g., classes 2 and 3)
- Client 2: Gets 2 shards (e.g., classes 4 and 5)
- Client 3: Gets 2 shards (e.g., classes 6 and 7)
- Client 4: Gets 2 shards (e.g., classes 8 and 9)

Why "Pathological"?
The term "pathological" refers to the extreme nature of this split:

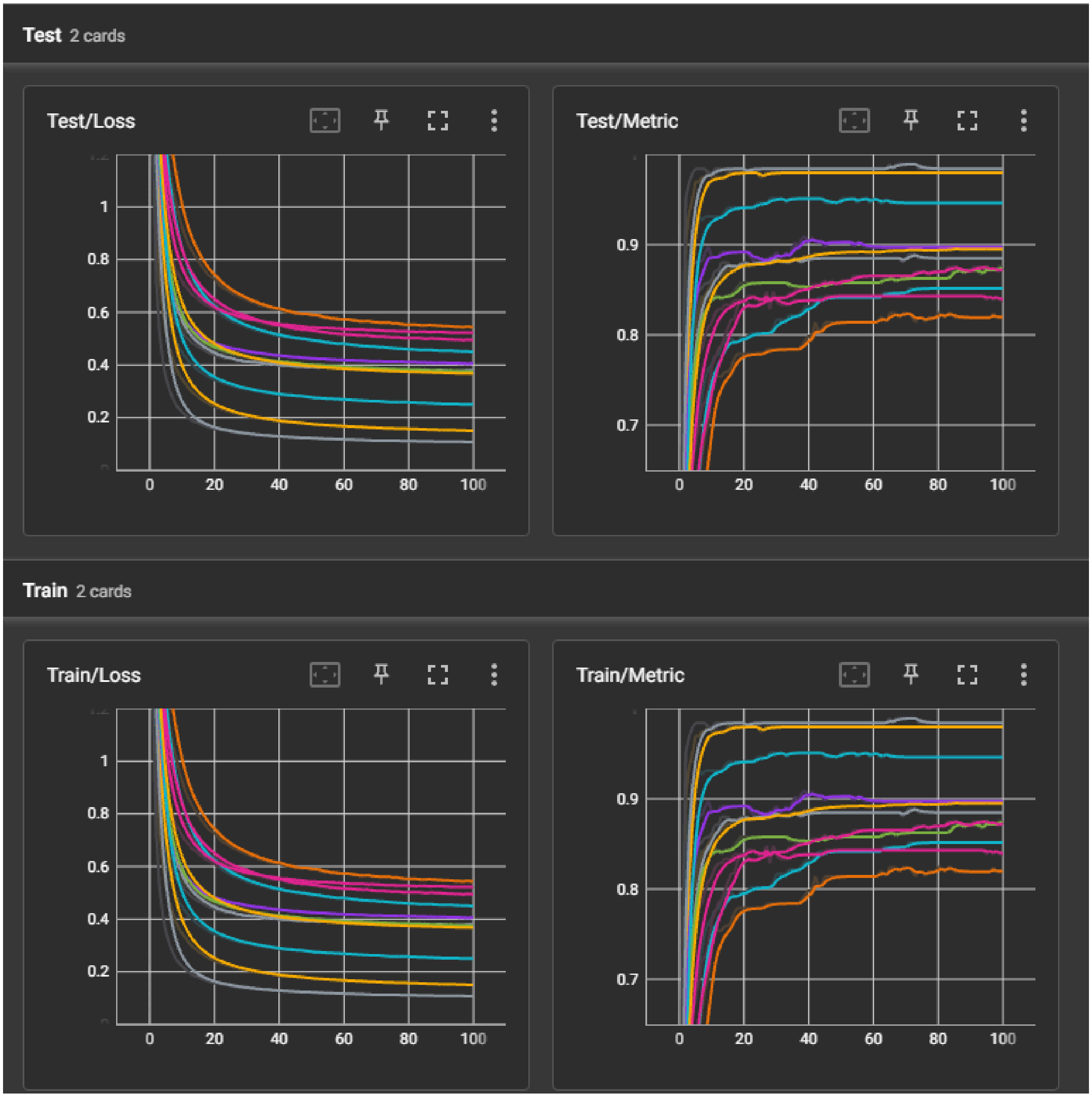
- **Severe class imbalance:** Each client sees only a tiny fraction of all classes
- **Maximum heterogeneity:** Creates the most challenging non-IID scenario
- **Realistic worst-case:** Represents extreme but realistic scenarios (e.g., specialized devices, geographic restrictions)

Plot: Experiments to observe how varying the number of local epochs (e.g., 1, 5, 10) influences the model's test accuracy under non-IID data distribution.
Plot the relationship between the number of local epochs and test accuracy.

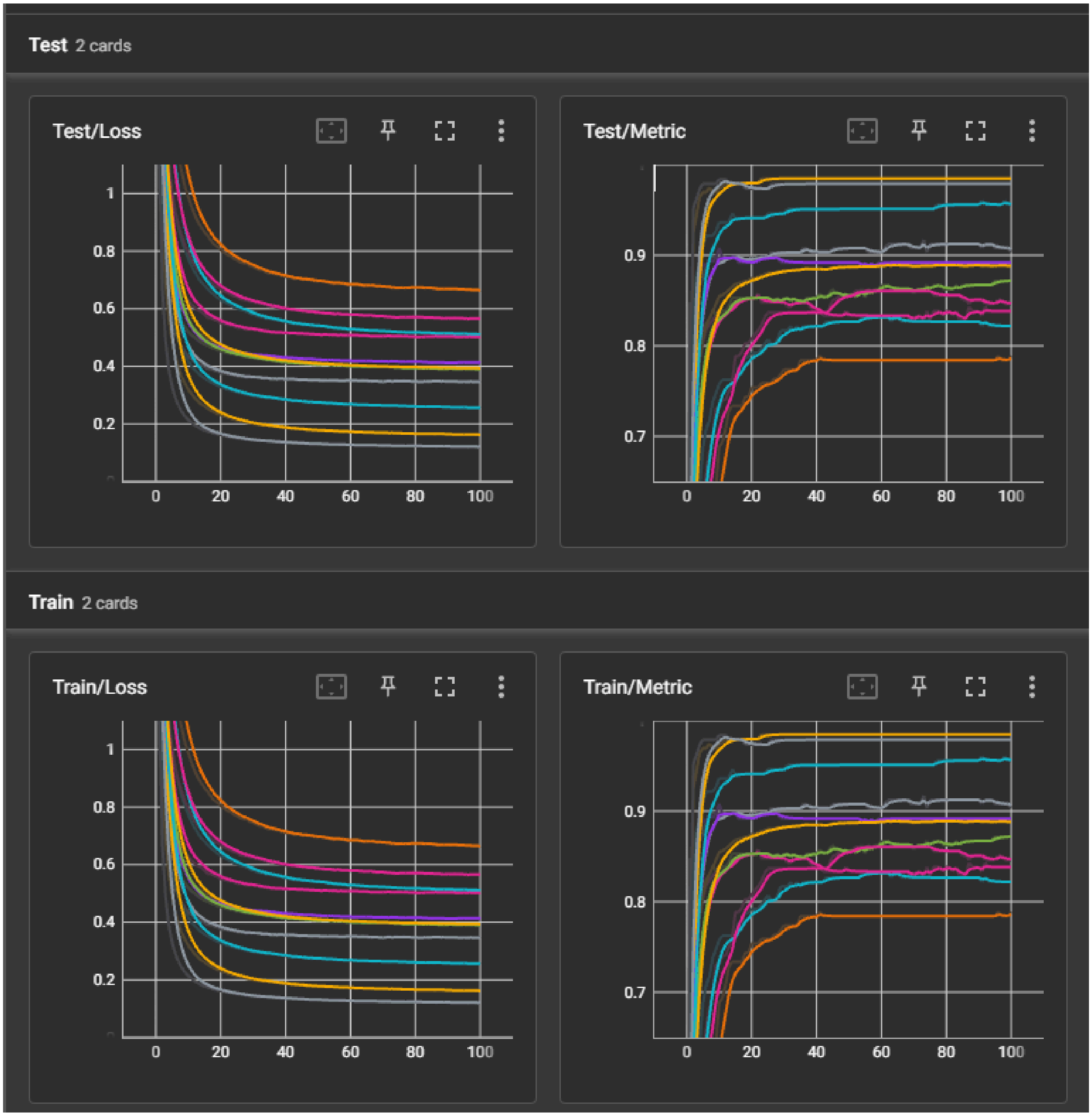
1. **Non-IID with local steps 1:**



2. **Non-IID with local steps 5:**



3. **Non-IID with local steps 10:**



Interpretation of Plots

As the number of local epochs increases, the model's test accuracy generally improves up to a point. When clients perform more local training steps before aggregation (e.g., from 1 → 5 → 10 epochs), they better optimize their local models, leading to faster convergence and higher global accuracy.

However, in a non-IID setting, this improvement is not strictly linear. After a certain number of local epochs, performance can plateau or even degrade slightly because:

Each client's model becomes more specialized to its own biased local data, diverging from the global objective.

The global aggregation struggles to reconcile these highly personalized updates, increasing variance in model parameters.

Overall, the results are expected:

- **Few local epochs (e.g., 1) :** slower convergence, lower accuracy.
- **Moderate local epochs (e.g., 5–10) :** improved accuracy and faster learning.
- **Too many local epochs (e.g., 50–100) :** potential overfitting to local data and instability in global performance, especially under non-IID distributions

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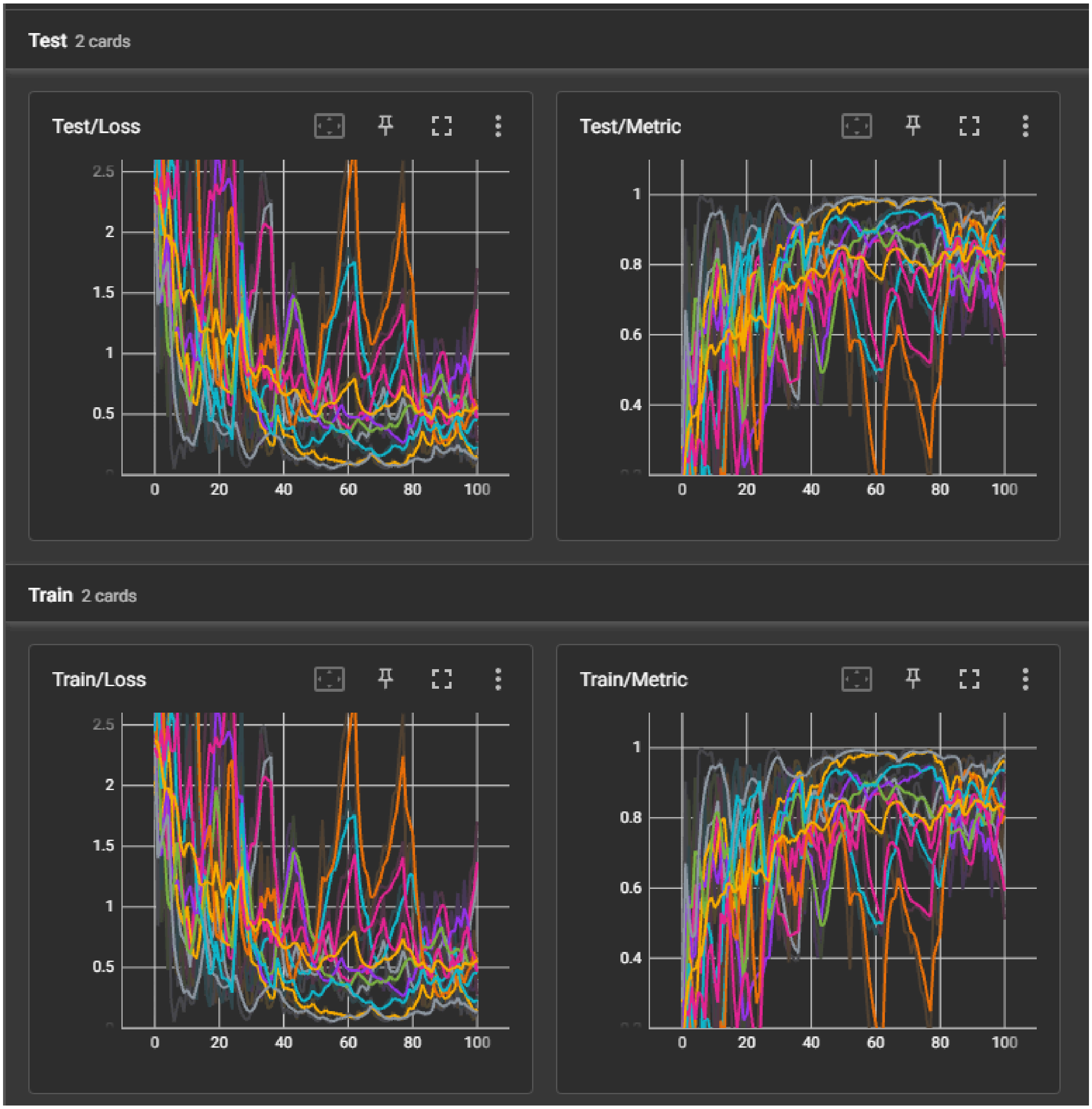
EXERCISE 5 - Client Sampling

EXERCISE 5.1 - Uniform Sampling Without Replacement



Explanation of Results

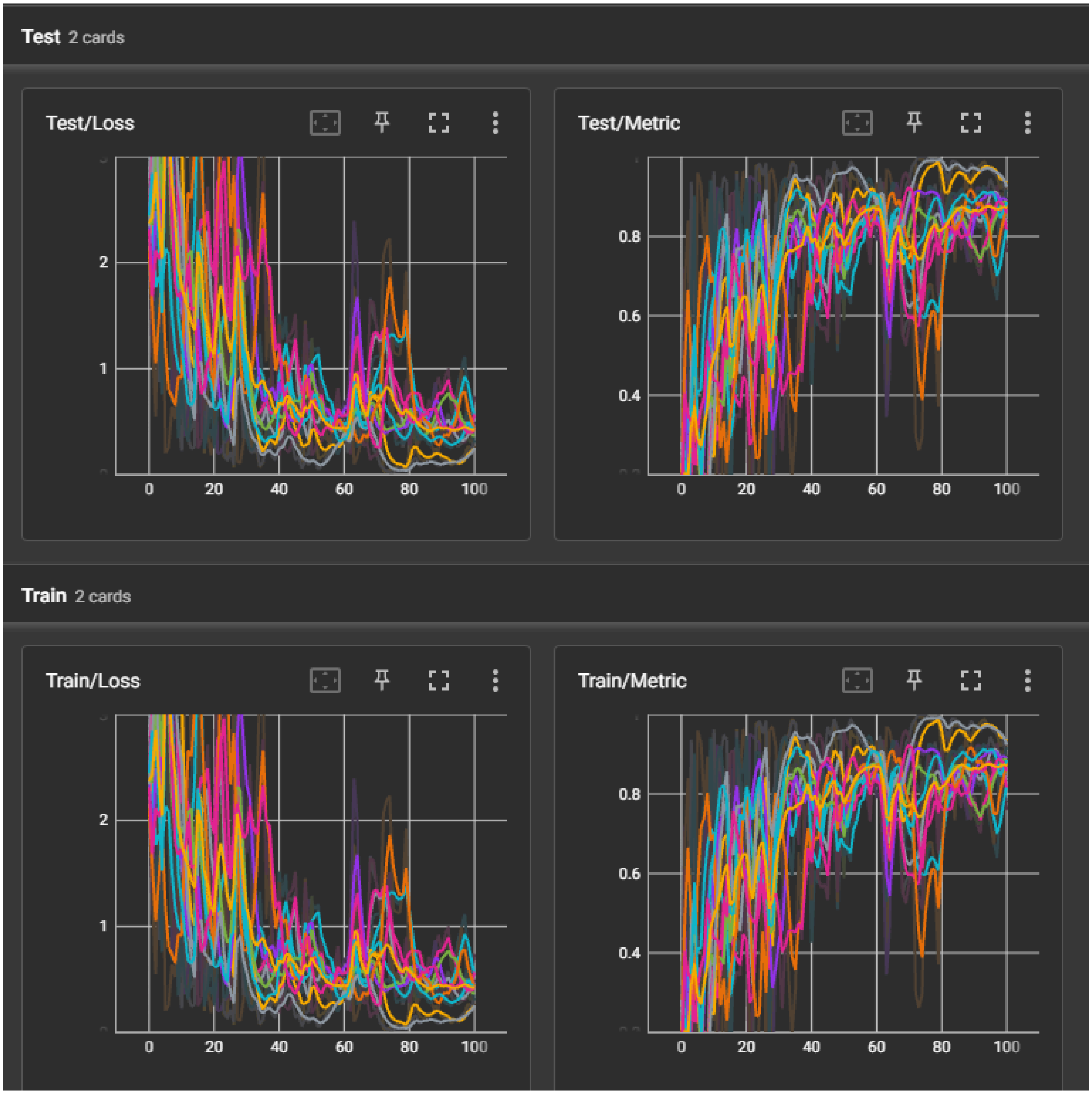
The plots display the evolution of training and testing loss and accuracy (metric) over communication rounds when client sampling with a rate of 0.2 was applied. Since only a random subset of clients (20% per round) participates in each aggregation step, the training dynamics show higher variability compared to the full-participation case.

- **Loss Behavior:**
  Both training and testing loss curves exhibit noticeable oscillations, particularly in early rounds, due to the reduced and varying number of clients contributing per round. This stochasticity arises because each sampled subset may represent different class distributions under non-IID data. Despite the fluctuations, the overall trend shows a gradual decrease in loss, indicating convergence.

- **Accuracy Behavior:**
  The training and testing accuracy (metric) curves steadily increase and stabilize near high values (~0.9–1.0) after several rounds, though they fluctuate more than in the non-sampled scenario. This confirms that even with partial participation, the FedAvg algorithm can reach satisfactory performance when the sampling rate is adequate (0.2 in this case).

- **Interpretation:**
  The results align with theoretical expectations. Uniform sampling without replacement introduces randomness but reduces communication cost. Although convergence is noisier and may be slightly slower, the model still achieves strong global performance. This demonstrates the trade-off between **communication efficiency** and **training stability** in federated learning under non-IID conditions.

EXERCISE 5.2 - Sampling With Replacement



Explanation of Results

The plots depict the training and testing loss and accuracy across communication rounds when client sampling with replacement was applied, using a sampling rate of 0.2. Compared to uniform sampling without replacement, the curves exhibit **slightly higher variability** across rounds. This is expected, as clients can be selected multiple times or not at all in a given round, introducing greater stochasticity into the aggregation process.

- **Loss Behavior:**
  Both training and testing loss decrease overall, showing convergence over time.

However, minor fluctuations persist throughout the rounds due to repeated selection of certain clients, which biases updates toward specific local data distributions. Despite this, the loss curves maintain a downward trend, indicating stable global learning.

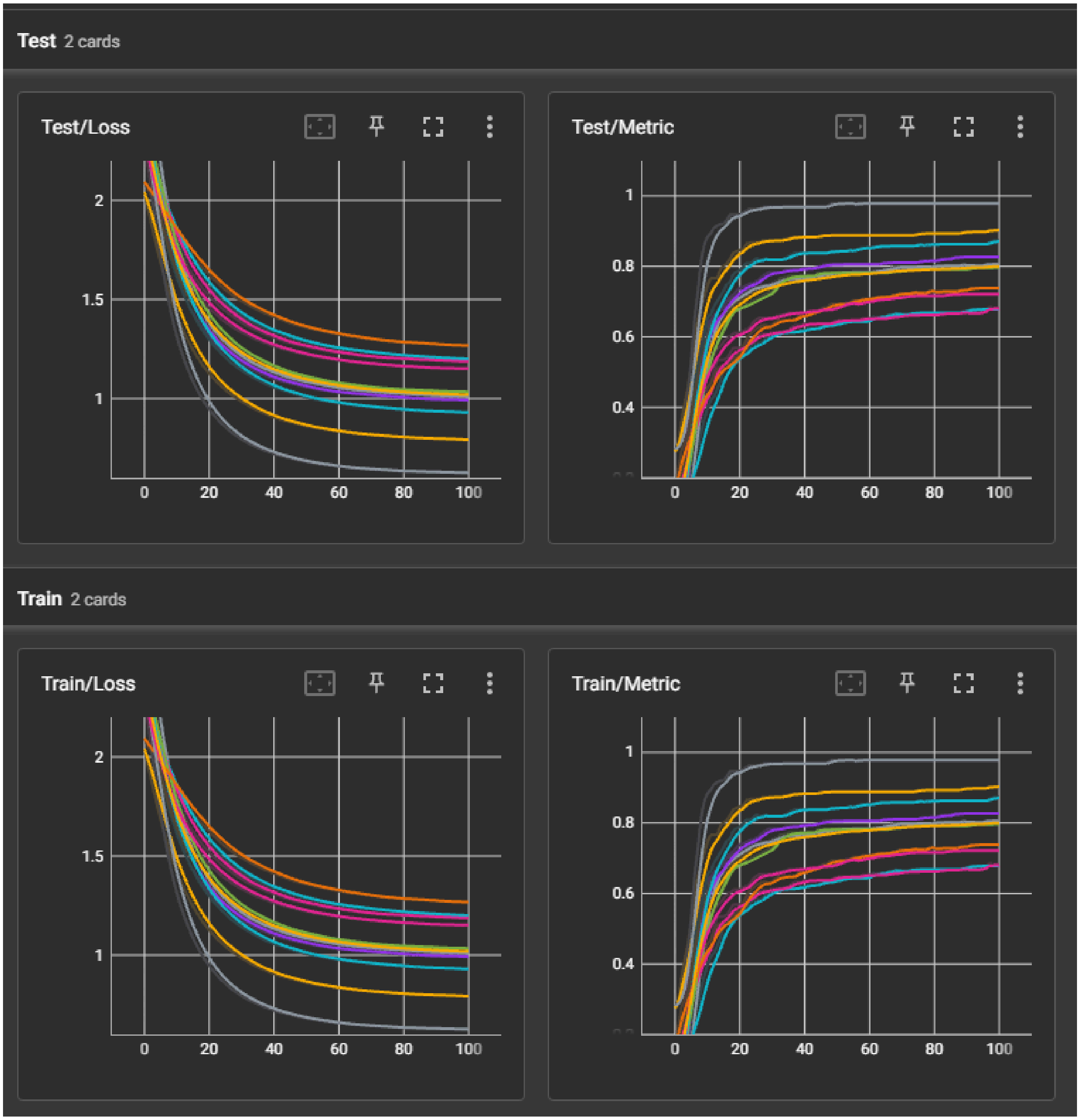
- **Accuracy Behavior:**
  Training and testing accuracy improve steadily and reach values close to 0.9–1.0, demonstrating effective learning. The fluctuations between rounds reflect the randomness of sampling, but the global model remains robust and convergent.

- **Analytical Insight:**
  Sampling with replacement enhances randomness and may slightly reduce convergence stability, but it provides a more flexible and unbiased estimate of the global model, particularly when client participation probabilities differ. The overall results are consistent with theoretical expectations: **while training becomes noisier, the algorithm still converges to a comparable accuracy**, confirming the resilience of FedAvg under probabilistic client participation.

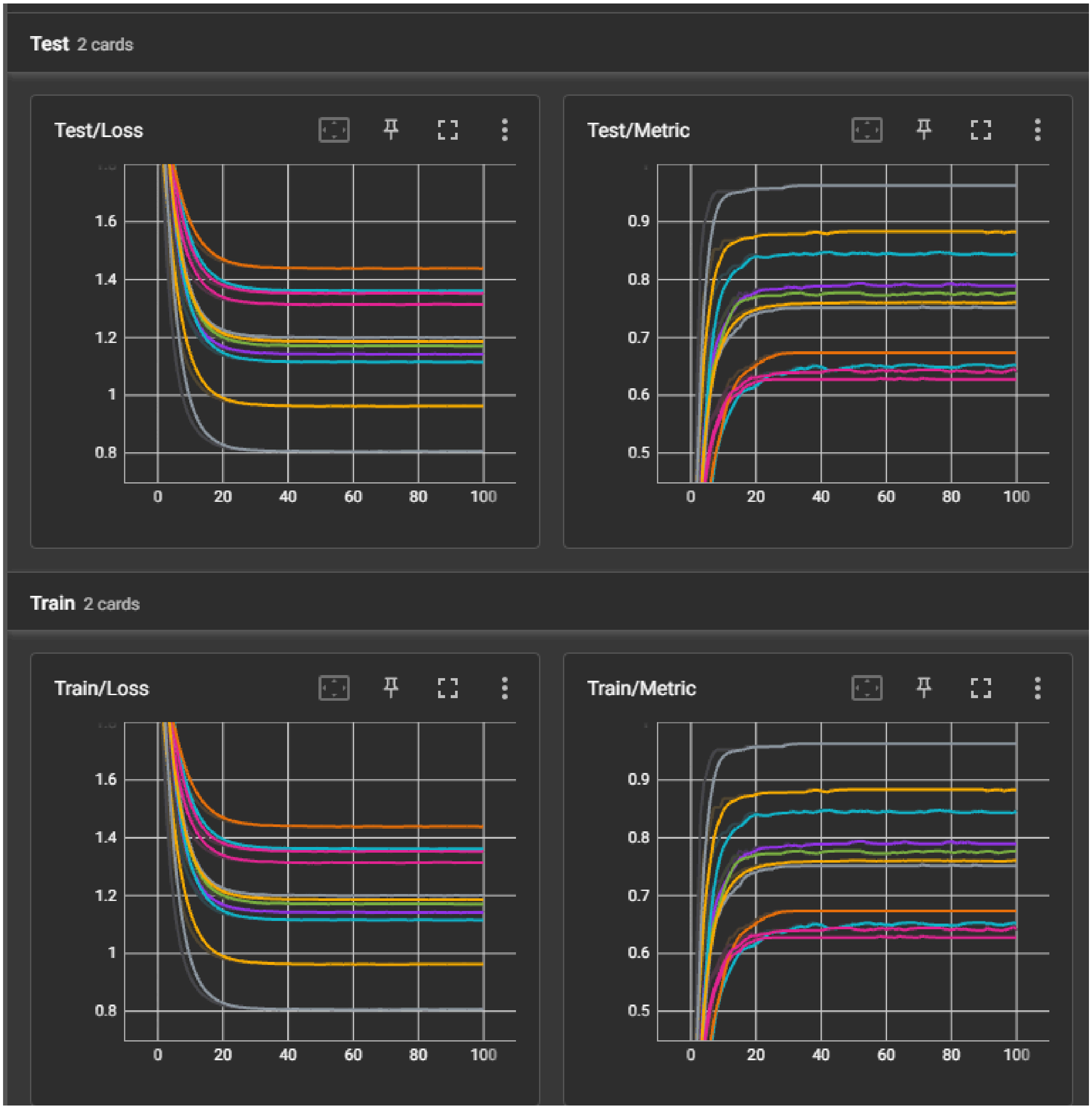
EXERCISE 6 - Algorithms

EXERCISE 6.1 - Tackling Data Heterogeneity with FedProx

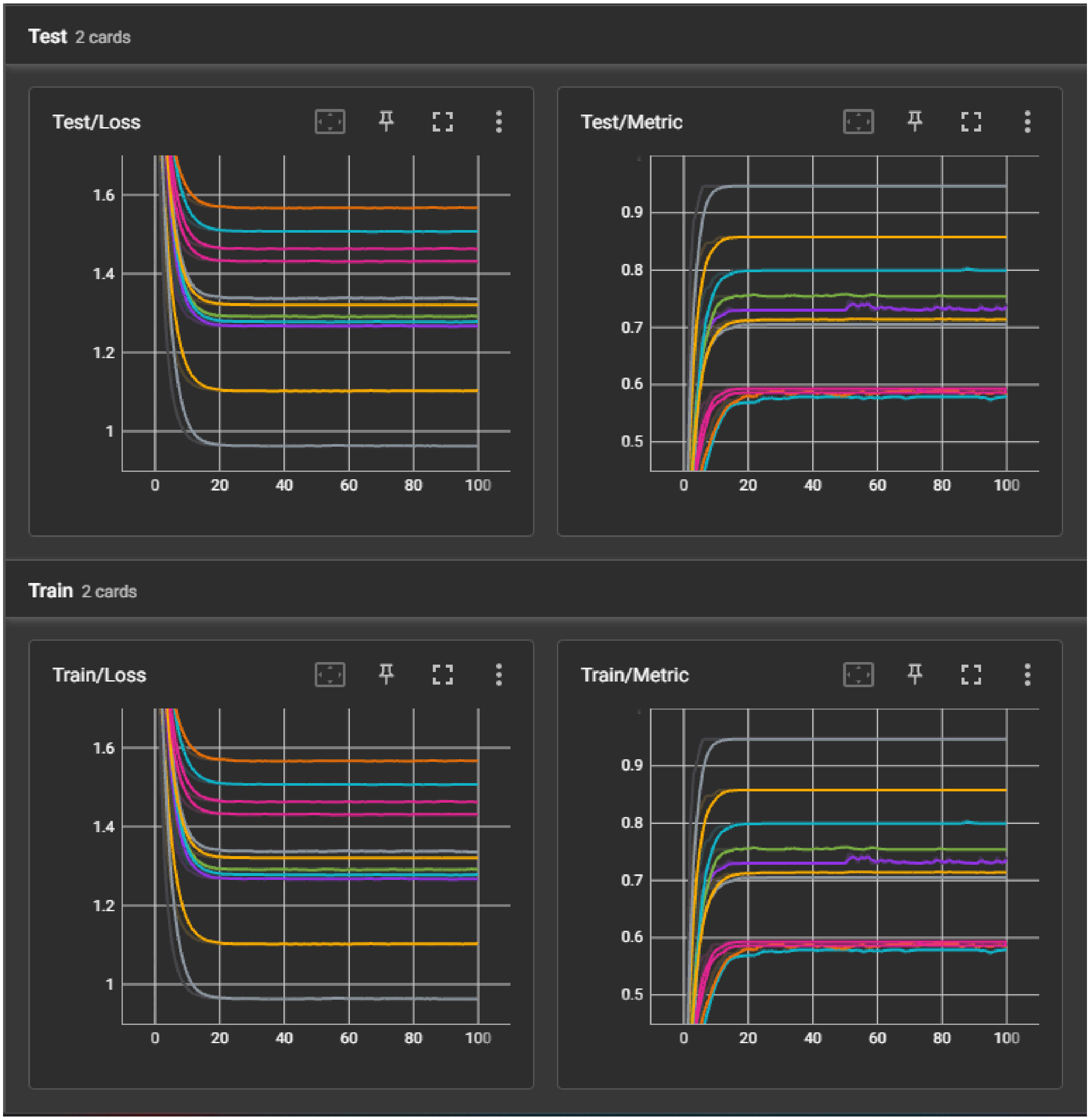
1. **FedProx with local steps 1**



2. **FedProx with local steps 5**



3. **FedProx with local steps 10**



Federated learning faces challenges when data across clients is non-IID, causing client drift where local models move toward their own biased optima. The FedAvg algorithm performs well in IID or low-epoch settings but struggles with heterogeneous data and many local updates, leading to unstable training and degraded accuracy. FedProx addresses this by introducing a proximal regularization term that constrains local updates to remain close to the global model, improving stability and convergence. As the number of local epochs increases, FedProx significantly outperforms FedAvg, especially under high data heterogeneity, partial client participation, or unbalanced datasets. While FedAvg is sufficient for homogeneous, IID environments with few local updates, FedProx is preferred for real-world federated settings where data and system heterogeneity are common. It provides better accuracy, smoother training dynamics, and stronger global convergence at minimal computational cost.

Configuration	FedAvg Performance	FedProx Performance	Key Insight
Few Local Epochs (E = 1–5)	Stable and accurate	Similar performance	Both perform equally when drift is minimal
Medium Local Epochs (E = 10)	Slight degradation, more oscillations	Stable and smooth convergence	Proximal term limits client drift
Many Local Epochs (E = 50–100)	Accuracy drops, unstable training	Robust accuracy, steady improvement	FedProx clearly superior under long local training
High Data Heterogeneity (Non-IID)	Divergent updates, inconsistent model	Controlled updates, better global accuracy	FedProx mitigates client drift effectively
Partial Client Participation	Sensitive to missing clients	More stable aggregation	FedProx tolerates system heterogeneity better
Unbalanced Data / High Learning Rate	Prone to overfitting or divergence	Regularization stabilizes updates	FedProx improves fairness and stability
IID, Few Epochs, Full Participation	Performs optimally	Matches FedAvg	FedAvg is sufficient in homogeneous conditions